

Summit Regenerative Orthopedics

Biologics-First. Non-Operative. Built for You.

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Regenerative Medicine 101 A Complete Patient Guide

Prepared for patients of Summit Regenerative Orthopedics | September 2026

Welcome — What Is Regenerative Medicine?

Regenerative medicine is a modern approach to healing that works *with* your body rather than around it. Instead of masking pain with medications or bypassing damaged tissue with surgery, regenerative medicine harnesses your body's own healing intelligence — and, where appropriate, carefully selected biological therapies — to repair, restore, and regenerate damaged tissue at its source. Whether the problem is a worn joint, a torn tendon, or chronic inflammation that simply will not resolve, regenerative therapies give your body the precise biological signals and building blocks it needs to do what it was always designed to do: heal. At Summit Regenerative Orthopedics, this philosophy is not just a treatment option — it is the foundation of everything we do.

Our Philosophy in One Sentence: Before we ever consider surgery, we ask

whether biology can do the job better — and in the vast majority of our patients, it can.

How Your Body Heals

To understand why regenerative medicine works, it helps to understand how your body naturally repairs itself — and why that process sometimes falls short.

The Basics of Tissue Repair

Any time you injure a muscle, tendon, ligament, or joint, your body launches an automatic healing response. This involves several key biological actors:

- **Growth Factors:** Signaling proteins that tell cells to multiply, migrate to the injury site, and begin rebuilding. Think of them as the body's construction foremen — they direct traffic and coordinate repairs.
- **Collagen:** The primary structural protein in tendons, ligaments, cartilage, and skin. After injury, the body produces new collagen fibers to patch the damaged area. The quality and organization of this new collagen determines how strong and functional the repaired tissue will be.
- **Inflammation:** Often misunderstood, acute inflammation is actually a necessary part of healing — it recruits repair cells to the injury site. The problem arises when inflammation becomes *chronic*, lingering long after it is useful and actually degrading the tissue it was meant to protect.
- **Platelets:** Tiny blood cells that rush to a wound site, form a clot, and release a concentrated burst of growth factors to initiate healing.

- **Stem and Progenitor Cells:** Undifferentiated cells that can transform into the specific cell types needed for repair — cartilage cells, tendon cells, bone cells, and more.

Why Healing Slows Down

Two factors significantly impair your body's natural repair capacity:

Factor	What Happens	The Result
Aging	Growth factor production declines; stem cells become less active; collagen quality decreases	Slower, weaker, incomplete tissue repair
Poor Blood Supply	Tendons, cartilage, and menisci have limited vascularity — few blood vessels to deliver repair cells	These tissues heal poorly on their own, even in younger patients
Chronic Inflammation	Persistent inflammatory chemicals break down cartilage and tissue faster than the body can rebuild	Progressive degeneration despite rest and conventional treatment
Repetitive Stress	Ongoing mechanical load outpaces the repair cycle	Micro-tears accumulate into chronic conditions like tendinopathy

This is precisely where biologics come in. Regenerative therapies replenish the growth factors, collagen precursors, and cell-signaling molecules that your body is no longer producing in sufficient quantities — essentially restarting or amplifying a stalled healing process.

The Cortisone Dilemma: Why We Don't Lead With Steroids

For decades, corticosteroid (cortisone) injections have been a first-line response to joint pain in conventional orthopedic care. They are fast, they are familiar, and — for a few weeks — they often work. But a growing body of peer-reviewed research reveals a troubling tradeoff: repeated corticosteroid injections may accelerate the very joint destruction they are meant to relieve.

Key Finding: Peer-reviewed studies published in journals including *Radiology* (Kompel et al., 2019) and the *Journal of Bone and Joint Surgery* have documented accelerated cartilage loss, subchondral bone changes, and increased rate of joint replacement in patients receiving repeat corticosteroid injections compared to those who did not.

What the Research Shows

- **Chondrocyte apoptosis:** Corticosteroids have been shown to directly trigger programmed cell death in chondrocytes — the cartilage-producing cells in your joints. Once chondrocytes are lost, cartilage cannot regenerate on its own.
- **Accelerated cartilage volume loss:** MRI studies tracking repeat steroid injection patients found measurably greater cartilage thinning over 1–2 years compared to control groups.
- **Weakening of periarticular structures:** Repeated exposure to corticosteroids weakens tendons, ligaments, and joint capsule tissue — increasing long-term instability risk.

- **Transient relief, progressive damage:** The anti-inflammatory effect masks pain effectively for 4–8 weeks, but does not address — and may worsen — the underlying structural deterioration.

The Biological Alternative

At Summit Regenerative Orthopedics, we do not use corticosteroid injections as a treatment modality. Instead of suppressing the inflammatory environment (and the tissue along with it), our biological therapies are designed to resolve inflammation appropriately and initiate actual tissue repair. For patients who have already received multiple cortisone shots, regenerative therapy may still be effective — though the timeline and response may differ. We will always give you an honest assessment.

If you have been told to “just get another cortisone shot,” we encourage you to ask what that shot is doing to your cartilage — and to explore whether a regenerative option might address the root cause instead.

Treatment Overview

Summit Regenerative Orthopedics offers a curated portfolio of evidence-supported biological therapies. Every recommendation is made based on your clinical picture alone — never on cost. In fact, our pricing is structured so that the same investment applies whether you receive PRP or a more advanced biologic like Wharton's Jelly or Exosomes. You will always receive the treatment that is right for *you*.

Treatment A Wharton's Jelly (Umbilical Cord-Derived)

What Is It?

Wharton's Jelly is a gelatinous substance found in the umbilical cord that surrounds and protects the cord's blood vessels. It is extraordinarily rich in the very biological components that support tissue repair and regeneration. Wharton's Jelly products used in clinical practice are derived from ethically donated umbilical cords — obtained with full informed consent from healthy mothers following planned cesarean deliveries, with no harm to mother or baby. The cord would otherwise be discarded.

What Does It Contain?

- **Mesenchymal Stem Cells (MSCs):** Multipotent cells capable of differentiating into cartilage, bone, tendon, and muscle tissue
- **Growth Factors:** A rich concentration of signaling proteins (including TGF- β , VEGF, IGF) that direct cell recruitment and proliferation
- **Collagen (Types I, III, and IV):** Ready-to-use structural proteins to scaffold new tissue growth
- **Hyaluronic Acid:** A natural lubricant and shock absorber particularly important for joint health
- **Cytokines:** Immune-modulating molecules that help regulate inflammation and create an environment conducive to healing
- **Exosomes:** Cell-signaling nanoparticles (see below) naturally present within Wharton's Jelly

How Is It Administered?

Wharton's Jelly is delivered via a targeted injection directly into the affected area — joint space, tendon sheath, bursa, or soft tissue — guided by ultrasound or fluoroscopy when appropriate to ensure precision. The procedure is performed in-office and typically takes less than 30 minutes.

What Conditions Does It Help?

- Moderate to severe osteoarthritis (knee, hip, shoulder, ankle)
- Cartilage defects and degeneration
- Chronic tendon and ligament injuries
- Labral tears (hip and shoulder)
- Degenerative disc disease
- Post-surgical recovery support

Why Wharton's Jelly? Because neonatal tissue is biologically "young" — it carries a high concentration of active growth factors and cellular components that adult tissue simply cannot match. Patients who might not respond fully to PRP alone often experience significantly better outcomes with Wharton's Jelly.

Treatment B Exosomes

What Are They?

Exosomes are nanoscale extracellular vesicles — tiny biological "packets" released by cells to communicate with one another. Think of them as the body's cellular messaging system: they carry genetic instructions (messenger RNA, microRNA), proteins, and lipids from one cell to another, essentially telling recipient cells what to do and how to behave. At approximately 30-150 nanometers in diameter, they are far smaller than a cell but carry an outsized biological impact.

How Do They Work?

When exosomes are introduced into an injured or degenerated area, they deliver their cargo — growth signals, anti-inflammatory instructions, and regenerative genetic messages — directly into local cells. This reprograms damaged cells toward a repair and regeneration state. Unlike cells themselves, exosomes do not divide or

persist permanently; instead, they act as a powerful, targeted signaling catalyst that kick-starts the body's own healing machinery.

Why Are They Cutting-Edge?

Exosomes represent one of the most rapidly advancing frontiers in regenerative medicine. Because they are cell-derived but acellular (they contain no living cells themselves), they offer a uniquely targeted, low-risk signaling approach. Research continues to expand our understanding of their role in tissue regeneration, immune modulation, and anti-inflammatory pathways.

What Conditions Do They Help?

- Degenerative joint conditions
- Chronic inflammation that has been resistant to other treatments
- Tendinopathy and soft tissue injuries
- Post-procedural recovery acceleration
- Conditions requiring targeted cellular reprogramming

Exosomes and Wharton's Jelly together: These two therapies are frequently used in combination. Wharton's Jelly provides structural and cellular building blocks, while concentrated exosome preparations amplify the signaling environment. Your provider will determine whether this combination is clinically appropriate for your condition.

Treatment C Platelet-Rich Plasma (PRP)

What Is It?

Platelet-Rich Plasma, or PRP, is a concentration of your own platelets and growth factors derived from a small sample of your blood. It is the most established and

widely studied of our regenerative therapies, with decades of clinical research supporting its use across orthopedic conditions.

The Process

1. **Blood Draw:** A small amount of your blood (typically 15–60 mL, similar to a routine lab draw) is collected from your arm.
2. **Centrifugation:** The blood is placed in a centrifuge, which spins it at high speed to separate red blood cells, white blood cells, and plasma. The platelet-rich layer is isolated.
3. **Concentration:** The resulting PRP contains a platelet concentration 3–8 times greater than normal blood, delivering a potent burst of growth factors to the treatment site.
4. **Injection:** The PRP is injected precisely into the injured area, triggering a targeted healing response.

How Do Platelets Help?

Platelets are nature's first responders to tissue injury. When activated, they release a cascade of growth factors — including Platelet-Derived Growth Factor (PDGF), Transforming Growth Factor-Beta (TGF- β), and Vascular Endothelial Growth Factor (VEGF) — that recruit repair cells, stimulate collagen production, promote new blood vessel formation, and modulate inflammation. PRP essentially delivers a concentrated version of this natural response exactly where it is needed.

What Conditions Does PRP Excel At?

- **Tendon injuries:** Patellar tendinopathy, Achilles tendinopathy, rotator cuff tendinopathy, tennis/golfer's elbow (lateral/medial epicondylitis)
- **Ligament sprains and partial tears:** MCL, LCL, plantar fascia
- **Mild to moderate osteoarthritis:** Particularly effective in early joint degeneration
- **Muscle strains** and soft tissue injuries
- **Cartilage support** in combination with other therapies

Remember: At Summit Regenerative Orthopedics, PRP is priced the same as our advanced biologics. If PRP is the best clinical fit for your condition, that is what we recommend — not because it costs less, but because it is the right tool for the job.

Treatment D MSK Laser (Musculoskeletal Laser Therapy)

What Is It?

MSK Laser therapy — also known as photobiomodulation or low-level laser therapy (LLLT) — uses specific wavelengths of light energy to penetrate tissue and stimulate healing at the cellular level. Unlike surgical lasers that cut or ablate tissue, therapeutic lasers deliver non-thermal, non-damaging light that interacts with cellular structures to produce measurable biological effects.

How Does It Work?

The primary mechanism involves the mitochondria — the energy-producing organelles within each cell. Laser light at therapeutic wavelengths (typically 600–1000 nm) is absorbed by cytochrome c oxidase, a key enzyme in the mitochondrial respiratory chain. This absorption:

- Increases the production of adenosine triphosphate (ATP), the cell's primary energy currency — effectively energizing cells to perform repair functions more efficiently
- Reduces oxidative stress within inflamed tissue
- Modulates inflammatory pathways, reducing pro-inflammatory cytokines
- Stimulates collagen synthesis and fibroblast activity
- Promotes local circulation and lymphatic drainage
- Reduces pain signaling by modulating nerve conduction

How Is It Used at Summit?

MSK Laser is valuable both as a standalone treatment for acute inflammation and as a powerful complement to injection-based biologics. When used after a PRP or Wharton's Jelly injection, laser therapy can enhance the local healing environment, reduce post-injection soreness, and help sustain the regenerative cascade initiated by the biologic. Treatment sessions are quick, comfortable, and entirely non-invasive.

Conditions Treated

- Acute and chronic musculoskeletal pain
- Tendinopathies and bursitis
- Joint inflammation
- Wound healing and scar tissue remodeling
- Nerve pain and neuropathy
- Adjunctive support following regenerative injections

Treatment E Peptides

What Are Peptides?

Peptides are short chains of amino acids — the same building blocks that make up proteins — that act as biological signaling molecules throughout the body. They are naturally occurring in human physiology and play roles in virtually every system: tissue repair, inflammation regulation, immune modulation, hormone signaling, and more. In a clinical context, specific therapeutic peptides are selected for their targeted effects on healing, recovery, and tissue regeneration.

How Do They Support Healing?

Therapeutic peptides relevant to musculoskeletal health work through several mechanisms:

- **Stimulating growth hormone and IGF-1:** Certain peptides (such as CJC-1295 or Ipamorelin) support the body's production of growth hormone, which plays a central role in tissue repair, muscle preservation, and collagen synthesis.
- **Promoting angiogenesis:** Some peptides encourage new blood vessel formation, improving nutrient and oxygen delivery to injured tissue.
- **Accelerating soft tissue healing:** BPC-157 (Body Protection Compound) has demonstrated a remarkable ability to accelerate tendon, ligament, and muscle repair in research models.
- **Reducing systemic inflammation:** Peptides can modulate immune responses to create a systemic environment more conducive to healing.
- **Supporting collagen production:** Peptides signal fibroblasts to produce more collagen, improving the structural quality of repaired tissue.

How Do They Complement Injection Therapies?

Peptide therapy works at a systemic level, creating a body-wide healing environment that amplifies the local effects of injected biologics. A patient receiving Wharton's Jelly for knee osteoarthritis may benefit significantly from concurrent peptide support that enhances their overall regenerative capacity, preserves surrounding muscle, and accelerates tissue remodeling throughout the recovery period. Peptides are typically administered via subcutaneous injection (similar to an insulin pen) and may be incorporated into your personalized treatment plan.

Personalized Peptide Selection: The specific peptides appropriate for you depend on your condition, health history, and goals. Your provider will discuss which peptides, if any, make sense as part of your individualized plan.

Why Biologics Before Surgery?

At Summit Regenerative Orthopedics, our providers come from a surgical background — which means when we recommend against surgery, we know exactly what we are recommending against. Surgery is sometimes the right answer. But for the majority of musculoskeletal conditions, it should not be the first answer.

The Case for a Biologics-First Approach

Consideration	Surgery	Biologics-First
Recovery time	Weeks to months of restricted activity, physical therapy	Minimal to no downtime; most patients resume activity quickly
Risk profile	Anesthesia risks, infection, scarring, nerve damage, hardware complications	Very low risk; primarily minor injection-site soreness
Root cause	Removes, repairs, or replaces damaged tissue mechanically	Stimulates the body to repair and regenerate tissue biologically
Future options	Surgery alters anatomy — future options may be limited	Biologics preserve anatomy; surgery remains an option if needed
Systemic impact	Significant physiological stress; immune, metabolic burden	Minimal systemic impact; may actually improve systemic health

What Our Non-Operative Philosophy Means for You

As a non-operative orthopedic practice, we approach every patient with the goal of finding the most effective path to healing that does not require going to an operating room. This means:

- A thorough evaluation of your imaging, history, and functional status before any recommendation is made

- An honest conversation about whether your condition is appropriate for regenerative treatment
- No pressure toward any particular product or treatment — only what the evidence and your clinical picture support
- A commitment to monitoring your progress and adjusting the plan if needed

Many of our patients have been told by other providers that surgery is their only option. We frequently find that a well-designed biological treatment plan can deliver meaningful, lasting relief — restoring function and quality of life without an operating room. **You deserve to know all your options before making that decision.**

What to Expect: Your Patient Journey

We believe informed patients heal better. Here is what your experience at Summit Regenerative Orthopedics typically looks like, from your first contact to your follow-up care.

1	Initial Consultation You meet with your provider for a comprehensive evaluation of your history, symptoms, lifestyle, and goals. This is a conversation — we want to understand the full picture of your health and what matters most to you.
2	Imaging Review We review existing X-rays, MRI, or ultrasound imaging — and may order additional studies if needed. Precise diagnosis is the foundation of a precise treatment plan.

1	Initial Consultation You meet with your provider for a comprehensive evaluation of your history, symptoms, lifestyle, and goals. This is a conversation — we want to understand the full picture of your health and what matters most to you.
3	Customized Treatment Plan Your provider presents a personalized plan tailored to your specific diagnosis, severity, and goals. This may include one or more biological therapies, laser sessions, peptides, and supportive care recommendations. All questions are welcomed and answered.
4	Treatment Day Most procedures are performed in-office in under 30 minutes. You will be made comfortable, the area will be prepared, and the treatment will be administered with precision. You can typically drive yourself home and return to light activity the same day.
5	Recovery & Home Care You will receive specific post-treatment instructions — including any activity modifications, icing protocols, and supplements or medications to avoid. Many patients notice an initial flare of soreness in the first few days as the healing response activates. This is normal and expected.
6	Follow-Up & Monitoring We schedule check-ins to monitor your progress, assess your response, and make any adjustments to your plan. Regenerative medicine is a process — we track your journey with you.

Realistic Timeline for Results

Regenerative therapies work by stimulating a biological process — and biology takes time. Understanding this timeline helps set appropriate expectations:

Timeframe	What Patients Typically Experience
Days 1-5	Possible initial soreness or swelling at the injection site as the healing response activates. Some patients notice early improvement in inflammation; others feel temporarily worse before improving. Both are normal.
Weeks 2-6	The regenerative cascade is underway. Many patients begin noticing meaningful reduction in pain and improvement in function. PRP and laser patients often notice early benefits in this window.
Months 2-3	Tissue remodeling is progressing. Patients receiving Wharton's Jelly or Exosomes often experience their most significant improvements in this period as new tissue matures.
Months 4-6	Maximum benefit is typically achieved. Many patients report sustained, durable relief that continues to improve even after the active treatment period ends.

Important: Individual results vary based on diagnosis, severity, age, overall health, and compliance with post-treatment guidance. Your provider will give you personalized expectations based on your specific situation. We are always transparent about what is and is not likely to respond to biological treatment.

Frequently Asked Questions

Q: Is regenerative medicine covered by my insurance?

Most regenerative biological therapies — including PRP, Wharton's Jelly, and Exosomes — are currently considered elective by most insurance carriers and are not covered under standard plans. We believe in full price transparency and will provide you with a clear understanding of all costs before you commit to anything. We also offer flexible payment options and can discuss financing if needed. Many patients find that the cost of a single well-designed regenerative treatment is far less than the cumulative cost of ongoing pain management, physical therapy, and eventual surgical intervention.

Q: Are there any side effects or risks?

Regenerative therapies have an excellent safety profile. The most common side effects are temporary injection-site soreness, mild swelling, and a short-term inflammatory flare — all of which are signs that the healing response is activating and typically resolve within a few days. Because PRP uses your own blood, allergic reactions are virtually impossible. Wharton's Jelly and Exosome products are screened and processed under strict quality standards. Serious adverse events are rare. Your provider will review your personal health history for any individual considerations before treatment.

Q: How do I know which treatment is right for me?

That is exactly what your consultation is designed to determine. Your provider will evaluate the type of tissue involved, the severity and chronicity of your condition, your imaging findings, your overall health, and your goals — and then make a recommendation based on clinical merit alone. At Summit Regenerative Orthopedics, our pricing structure is intentionally designed so that cost is never a factor in what we recommend. You will always receive the most appropriate treatment for your condition.

Q: How long do results last?

For many patients, results are durable — lasting one to several years or longer, particularly when the underlying condition has been meaningfully addressed rather than simply managed. Conditions that are more advanced, or in environments of continued mechanical stress, may benefit from periodic maintenance treatments. PRP for tendinopathy tends to provide lasting results in properly selected patients. Advanced biologics like Wharton's Jelly often produce sustained benefits given the structural rebuilding they support. Your provider will give you realistic expectations based on your specific diagnosis.

Q: Can I combine treatments?

Absolutely — and in many cases, combination therapy produces superior outcomes. For example, Wharton's Jelly paired with concentrated Exosomes addresses both the structural and signaling aspects of tissue repair simultaneously. Adding MSK Laser therapy can enhance the local healing environment and reduce inflammation that might otherwise blunt the biologic's effect. Peptide therapy works at a systemic level to support the entire healing cascade. Your provider will design a plan that uses each therapy in the role it performs best.

Q: What if I have already had surgery on this area?

Post-surgical patients are excellent candidates for regenerative therapy in many cases. Surgery can leave behind scar tissue, altered mechanics, and residual inflammation that biologics can meaningfully address. We frequently work with patients who continue to experience pain or functional limitations following orthopedic surgery. Prior surgery does not disqualify you — it simply becomes part of the clinical picture we consider when designing your plan.

Q: Are the biologic products safe and properly tested?

Yes. All biologic products used at Summit Regenerative Orthopedics are sourced from FDA-registered tissue banks and laboratories that comply with Current Good Tissue Practice (cGTP) regulations. Wharton's Jelly and Exosome products undergo rigorous donor screening, infectious disease testing, and quality control testing before they are cleared for clinical use. We are selective about the partners we work with and will always be transparent about the sourcing and testing standards of the products we use.

Q: Do I need to stop any medications before treatment?

Some medications can interfere with the biological response we are trying to create. Most notably, anti-inflammatory medications (NSAIDs such as ibuprofen or naproxen) and corticosteroids can blunt the healing cascade that regenerative therapies depend on, and we typically ask patients to avoid these for a period before and after treatment. Blood thinners and certain supplements may also need to be managed. Your provider will give you a specific pre-treatment medication protocol at your consultation. Never stop a prescription medication without first discussing it with the prescribing physician.

Our Promise to You

At Summit Regenerative Orthopedics, we made a deliberate choice to build a practice around a single guiding belief: that most patients deserve a meaningful biological treatment option before they are ever handed a surgical consent form.

That belief shapes everything we do:

- **Biologics-first, always.** We do not recommend surgery when biology can do the job. We do not recommend advanced biologics when PRP will suffice. And we do not recommend PRP when watchful waiting is appropriate. Every recommendation is made with your best outcome as the only goal.
- **Same price, honest advice.** Our pricing is structured so that the cost of a treatment never influences what we recommend. PRP and Wharton's Jelly cost the same here — so when we recommend one over the other, you can trust it is because of your clinical picture, not our margin.
- **Non-operative by philosophy.** We are not anti-surgery — we are pro-options. We believe you deserve to know what your body is capable of healing before anyone picks up a scalpel. Our background in orthopedic surgery gives us the expertise to make that call with confidence.
- **Personalized to you.** There is no standard protocol here. Your age, anatomy, diagnosis, lifestyle, and goals all shape your treatment plan. You are not a diagnosis — you are a person, and we treat you accordingly.

Thank you for trusting us with your care. We look forward to being part of your healing journey.

— The Team at Summit Regenerative Orthopedics

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